

## Chtouka Aquifer Project — Setup & Run Instructions

This guide explains how to set up your computer and run the analysis notebook (Geospatial\_Analysis\_python\_script\_final.ipynb).

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### 1. Download the project

1. Go to the GitHub repository page.
  2. Click the green **Code** button → **Download ZIP** (or use `git clone <repo-url>` if you're comfortable with Git).
  3. Unzip it to a folder on your computer.
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### 2. Download the data

The dataset is too large to store on GitHub, so it's hosted separately.

1. Download the data package from this link:  
[\[https://drive.google.com/drive/folders/16vx7QMr9eQfzP6jOe\\_DjaKrue\\_1GLJ2G?usp=drive\\_link\]](https://drive.google.com/drive/folders/16vx7QMr9eQfzP6jOe_DjaKrue_1GLJ2G?usp=drive_link)
2. Unzip it.
3. Place the unzipped folder inside the project so it becomes a folder named exactly `data` at the top level of the project — right next to the notebook file.

Your project folder should look like this when done:

Geospatial\_Analysis\_Project/

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|
|— Geospatial_Analysis_python_script_final.ipynb
|— data/
|   |— Data_Chtouka_Reduced.csv
|   |— DataBaseChtoukaFinal_WGS/
|       |— Limite/
|       |— demf/
```

```
|   ├── LULC_10m_2024_2017/
|   ├── Costal_line/
|   ├── Desalinization_Network/
|   ├── FermesIrriguéesDessalemen/
|   ├── Irrigated_perimeters_Fig6&7/
|   └── ...
├── hydrorivers/
├── GeologyBuffer_new/
├── land_cover_cropped/
└── output/ (this will be created automatically when you run the script)
```

 Important: the folder must be named data (lowercase) and sit in the same location as the notebook. If it's named differently or placed elsewhere, the script won't find the files.